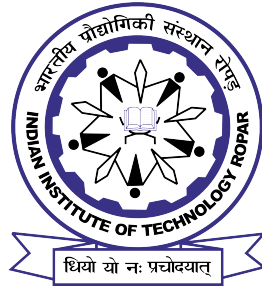

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Development Engineering Project

Decentralized Biogas Production with Centralized Upgrading & Grid Injection: When Is It Optimal?

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April 30, 2026

Abstract

The study conducts an exhaustive techno-economic analysis of biogas plant models, following the framework laid out by Hengeveld et al. It is carried out in three successive stages: validation of the model, correction and refinement of the model, and application of the modified model.

In the first phase, the original model was utilised to replicate results found in the literature and to validate its computational capacity. During the replication, problems in the parameters used, transportation distance, costing, and system configuration assumptions were identified.

During Phase 2, the errors in the original model were corrected and the model refined. This involved changing incorrect assumptions about some parameters and revising biomass supply and transportation cost figures to enhance accuracy, validity, and robustness of the model.

Phase 3 involves application of the modified model to a comparative study of the suitability of two regions in Rajasthan—Kota and Bikaner—for installation of biogas plants. Parameters for both cases were collected and analysed to determine suitability for decentralised biogas production with centralised upgrading and grid injection.

The conclusion drawn is that, although centralisation is linked to reduced cost of production through economies of scale, the evidence supports the idea of decentralisation plus centralisation relative to regional factors such as distance and availability of raw materials.

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Chapter 1

Aims and Objectives

The objectives of this study are:

1. To validate the original biogas production model.
2. To identify and correct errors and assumptions affecting model accuracy.
3. To apply the corrected model to real-world locations.
4. To determine the most suitable location and configuration for the biogas plant setup.

Chapter 2

Introduction

2.1 Background

Biogas production using anaerobic digestion is a proven technology for the conversion of organic biomass into renewable energy. This process entails microbial degradation whereby biodegradable materials such as animal manure and agricultural maize undergo decomposition in an oxygen-free environment. Biogas produced through anaerobic digestion contains mainly methane and CO_2 . The purified biogas can then be upgraded to become green gas or biomethane with properties similar to natural gas and injected into the natural gas distribution network.

Benefits of incorporating upgraded biogas into the natural gas network include easy transportation over long distances, convenient storage options, and compatibility with current infrastructure. Biogas technologies also help manage waste sustainably, reduce greenhouse gas emissions, and recycle nutrients via digestate as a fertiliser.

2.2 Centralized vs. Decentralized Systems

One of the most critical considerations in the design of biogas production systems is the choice of an optimal configuration. The two principal system types are:

- **Centralised systems**, in which biomass from different sources is transported to a large-scale digester, followed by upgrading and injection of biogas.
- **Decentralised systems**, in which biomass is sent to several smaller digesters, which produce biogas transported to a centralised upgrading point.

According to research by Hengeveld et al., a comprehensive operational model was developed for studying the whole supply chain, encompassing biomass production, transport of biomass and digestate, biogas transportation, digestion, and upgrading. Key findings include:

- Economies of scale in centralised systems reduce both capital and operating costs.
- Although centralised systems have higher biomass transport costs, decentralised systems incur higher costs from biogas transport.
- Total energy consumption per unit of green gas is approximately equal for both

configurations.

These findings indicate that there is no universal best solution; performance depends on an optimal balance between economies of scale and transportation constraints.

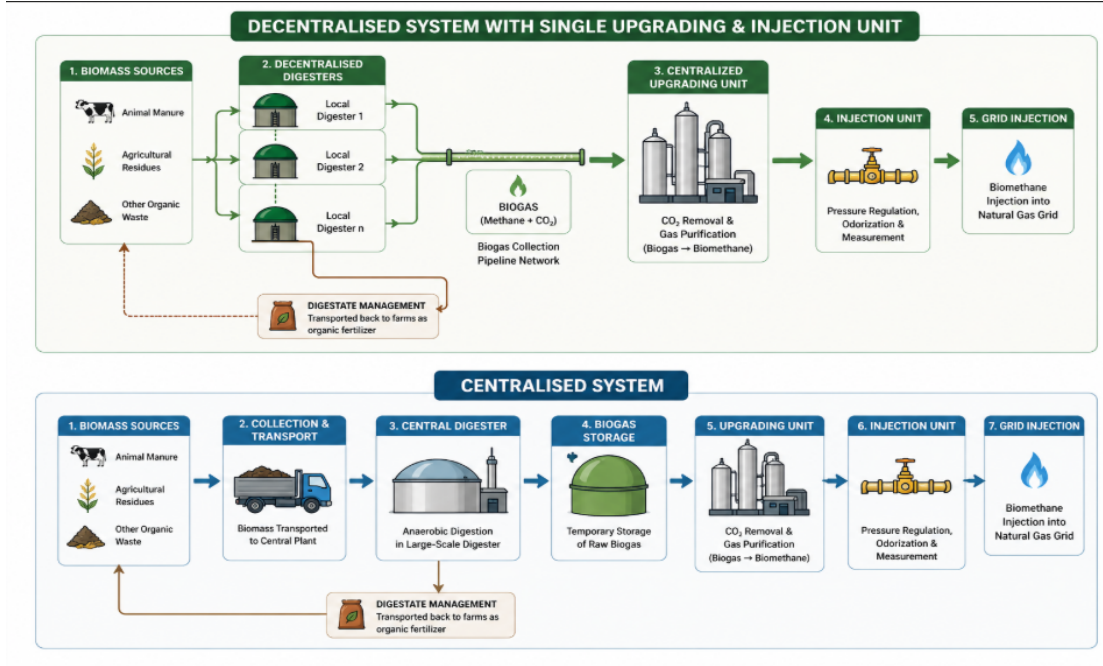


Figure 2.1: Schematic comparison of decentralised biogas production with a single centralised upgrading and injection unit and a fully centralised biogas production system.

2.3 Research Phases

To guarantee a systematic approach and reliable analysis, this project is structured in three consecutive steps.

2.3.1 Phase 1: Model Validation

The initial model suggested by Hengeveld et al. was implemented according to all assumptions, parameters, and configurations given in the reference publication. This phase included:

- Rewriting the entire biogas production process—biomass harvesting and delivery, processing, biogas transportation, upgrading, and injection.
- Conducting energy and cost calculations for centralised and decentralised configurations.
- Comparing output data with published values such as cost per unit of green gas and energy consumption.

Some errors related to outputs, misinterpretation of parameters, and implementation mistakes were identified.

2.3.2 Phase 2: Model Corrections and Enhancements

In light of problems found during Phase 1, the model was corrected and enhanced by:

- Correcting misinterpreted parameters, such as digester capacities, transportation distances, and cost coefficients.
- Revising assumptions regarding biomass availability, transportation, and configuration.
- Correcting errors in cost calculations, energy requirements, and result aggregation.
- Verifying that results exhibit expected trends such as economies of scale and configuration-dependent cost differences.

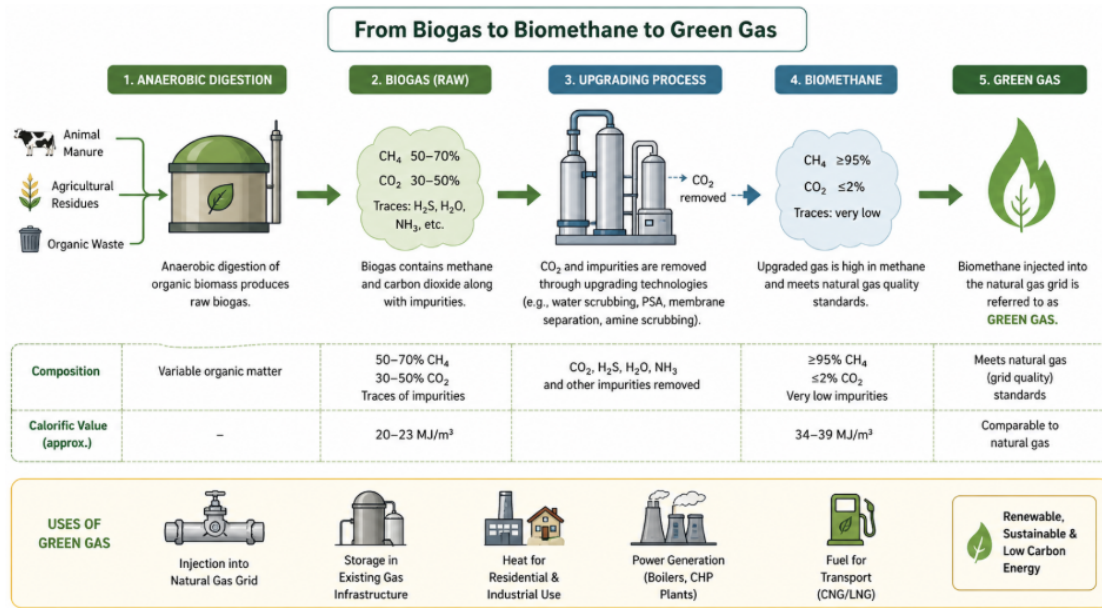


Figure 2.2: Process flow from raw biogas to biomethane or green gas, including anaerobic digestion, upgrading, and grid injection.

2.3.3 Phase 3: Location-Based Application

The corrected model was applied to a comparative analysis of Kota and Bikaner in Rajasthan. Factors considered include:

- **Biomass availability:** Amount and availability of animal manure and maize.
- **Logistics:** Distance, infrastructure, and ease of biomass and biogas transport.
- **Economics:** Capital and operational costs associated with system implementation.

Chapter 3

Terminology

Understanding key terminology is vital for discussions spanning chemical engineering, thermodynamics, fluid mechanics, and economics. The definitions below establish a common basis for comparing distributed and centralised configurations.

Biomass Renewable organic material derived from living or recently living organisms, primarily composed of carbon, hydrogen, oxygen, nitrogen, and trace alkaline earth metals.

Manure Animal excreta mixed with water and undigested feed residues. It serves as the steady base feed for digesters and contributes to daily digester loading and transport costs.

Co-substrate

Extra organic material, such as maize silage, mixed with manure to boost gas production. It improves nutrient balance for microorganisms and increases methane yield.

Anaerobic Digestion

A natural, oxygen-free biological process in which microorganisms break down organic materials into biogas and digestate.

Biogas A renewable combustible mixture comprising 50–70% methane, 30–40% carbon dioxide, and traces of hydrogen sulfide and moisture.

Green Gas or Biomethane

Upgraded biogas containing more than 95% methane, produced by removing carbon dioxide, hydrogen sulfide, and water vapour.

Digester A sealed vessel that maintains anaerobic conditions for microbial decomposition of organic waste into biogas.

Centralised Upgrading

A single large facility that purifies biogas collected from multiple dispersed digesters.

Biogas Grid

A network of pipes connecting distributed digesters to the centralised upgrading facility.

Pressure Drop

Reduction in gas pressure due to friction in pipelines, which requires pumping or compression energy.

Compressor

Equipment that increases gas pressure for efficient storage, transport, and utilisation.

Upgrading Process

A system that refines raw biogas into high-purity biomethane.

CAPEX Capital expenditure, or one-time investment cost, for equipment such as digesters, tanks, pipes, and storage infrastructure.

OPEX Operating expenditure, or ongoing cost, including labour, energy, and maintenance.

Specific Energy Consumption

Energy consumed per unit volume of green gas produced, expressed in MJ/m³.

Friction Factor

A dimensionless number representing resistance due to pipe-wall roughness and fluid viscosity.

Digestate

The nutrient-rich solid or liquid residue remaining after anaerobic digestion, which can be used as organic fertiliser.

Grid Injection System

Equipment that delivers cleaned biomethane to the gas distribution network after final pressure regulation, quality checks, and metering.

Table 3.1: Distinction between biomethane and green gas.

Aspect	Biomethane	Green Gas
Nature	Chemical composition	Functional or market term
Definition	Upgraded biogas	Grid-quality renewable gas
Focus	Purity and composition	Usability and application
Engineering interpretation	Product of upgrading process	Usable energy carrier in the grid

Chapter 4

Methodology

4.1 Overview

The entire model is implemented in an Excel spreadsheet named “Biogas Grid”, where each worksheet represents a process block. All parameter values are explicitly listed in each sheet, and step-by-step calculations are performed using those values. The final outputs are:

- Cost in €/m³ or €/t/m³.
- Energy consumption in MJ/m³.

4.2 Input Parameters

4.2.1 System Configuration

Table 4.1: System configuration input parameters.

Parameter	Value
Number of digesters	8
Biogas per digester	254.5 m ³ /year
Operating hours	8000 h/year
Annual green gas	9,600,000 m ³ /year

4.2.2 Feedstock

Table 4.2: Feedstock parameters.

Parameter	Value
Number of cows	4,064
Manure production	20 ton/cow/year
Co-substrate	205,000 ton/year
Co-substrate price	13.5 €/ton

4.2.3 Grid and Injection Parameters

Table 4.3: Grid and injection input parameters.

Parameter	Value
Pipeline radius	9.98 km
Pipeline diameter	110 mm
Pipeline cost	88 €/m
Injection distance	1450 m
Injection cost	130 €/m

4.3 Biomass Calculation

Manure Production

$$\text{Manure} = N_{\text{cows}} \times \text{Manure per cow} = 4,064 \times 20 = 81,280 \text{ ton/year} \quad (4.1)$$

Table 4.4: Total biomass input to the digester.

Stream	Quantity
Manure production	81,280 ton/year
Co-substrate	205,000 ton/year
Total biomass	286,280 ton/year

4.4 Biogas Production

Table 4.5: Biogas yield parameters and production calculation.

Feedstock	Production	Yield	Total Biogas
Manure	81,280 ton/year	25 m ³ /ton	2,032,000 m ³ /year
Co-substrate	205,000 ton/year	175 m ³ /ton	35,875,000 m ³ /year
Total biogas production			37,907,000 m³/year

The methane fraction is 0.556. Therefore:

$$\text{Methane production} = 37,907,000 \times 0.556 \approx 21,076,292 \text{ m}^3/\text{year} \quad (4.2)$$

4.5 Digester Cost and Energy

Table 4.6: Digester cost parameters.

Parameter	Value
Number of digesters	8
Investment per digester	2,025,000 €
Lifetime	12 years
OPEX fraction	5%

Step-by-step calculation:

$$\text{Total Investment} = 2,025,000 \times 8 = 16,200,000 \text{ €} \quad (4.3)$$

$$\text{Annual Depreciation} = \frac{16,200,000}{12} = 1,350,000 \text{ €/year} \quad (4.4)$$

$$\text{Annual OPEX} = 0.05 \times 16,200,000 = 810,000 \text{ €/year} \quad (4.5)$$

$$\text{Total Annual Cost} = 1,350,000 + 810,000 = 2,160,000 \text{ €/year} \quad (4.6)$$

$$\text{Cost per m}^3 = \frac{2,160,000}{9,600,000} = 0.225 \text{ €/m}^3 \quad (4.7)$$

Table 4.7: Digester energy consumption breakdown.

Component	Energy
Heating	1.194 MJ/m ³
Mixing	0.030 MJ/m ³
Pumping	0.015 MJ/m ³
Total	1.239 MJ/m³

Result: Cost = 22.5 €ct/m³; Energy = 1.239 MJ/m³.

4.6 Biomass Transport

4.6.1 Manure Transport

Table 4.8: Manure transport parameters.

Parameter	Value
Distance	10.6 km
Truck cost	1.24 €/km
Fuel energy	1.3 MJ/(ton km)
Truck capacity	30 ton

$$\text{Total Energy} = 81,280 \times 10.6 \times 1.3 = 1,121,029 \text{ MJ/year} \quad (4.8)$$

$$\text{Energy per m}^3 = \frac{1,121,029}{9,600,000} = 0.1167 \text{ MJ/m}^3 \quad (4.9)$$

$$\text{Truck trips} = \frac{81,280}{30} \approx 2,709 \text{ trips/year} \quad (4.10)$$

$$\text{Truck cost} = 2,709 \times 10.6 \times 1.24 \times 2 = 71,286 \text{ €/year} \quad (4.11)$$

$$\text{Cost per m}^3 = \frac{71,286}{9,600,000} = 0.007425 \text{ €/m}^3 \quad (4.12)$$

4.6.2 Maize or Co-substrate Transport

Table 4.9: Maize transport parameters for truck and tractor.

Parameter	Truck	Tractor
Distance	10.6 km	0.45 km
Cost	1.24 €/km	0.85 €/km
Fuel energy	1.3 MJ/(ton km)	2.1 MJ/(ton km)
Capacity	30 ton	14 ton

$$\text{Truck energy} = 205,000 \times 10.6 \times 1.3 = 2,827,399 \text{ MJ/year} \quad (4.13)$$

$$\text{Tractor energy} = 205,000 \times 0.45 \times 2.1 = 193,725 \text{ MJ/year} \quad (4.14)$$

$$\text{Energy per m}^3 = \frac{2,827,399 + 193,725}{9,600,000} = 0.3149 \text{ MJ/m}^3 \quad (4.15)$$

Table 4.10: Summary of biomass transport costs and energy.

Feedstock	Cost	Energy
Manure	0.7425 €/m ³	0.1167 MJ/m ³
Maize	1.9908 €/m ³	0.3149 MJ/m ³

4.7 Digestate Transport

$$\text{Total digestate} = 205,000 + 81,280 = 286,280 \text{ ton/year} \quad (4.16)$$

$$\text{Truck trips} = \frac{286,280}{30} \approx 9,543 \text{ trips/year} \quad (4.17)$$

$$\text{Ton-km} = 286,280 \times 10.61 = 3,037,253 \text{ ton km} \quad (4.18)$$

$$\text{Energy} = 3,037,253 \times 1.3 = 3,948,429 \text{ MJ/year} \quad (4.19)$$

$$\text{Transport cost} = 251,080 \text{ €/year} \quad (4.20)$$

$$\text{Energy per m}^3 = \frac{3,948,429}{9,600,000} = 0.411 \text{ MJ/m}^3 \quad (4.21)$$

$$\text{Cost per m}^3 = \frac{251,080}{9,600,000} = 2.615 \text{ €/m}^3 \quad (4.22)$$

4.8 Biogas Grid: Pipeline and Compression

Table 4.11: Biogas grid input parameters.

Parameter	Value
Radius	9.98 km
Diameter	110 mm
Cost	88 €/m
Total length	139.72 km

Pipeline cost:

$$\text{Pipeline investment} = 139,720 \times 88 = 12,295,360 \text{ €} \quad (4.23)$$

$$\text{Annual maintenance} = 245,907 \text{ €/year} \quad (4.24)$$

Additional costs: Desulfurisation and drying cost is 5000 € per digester, giving a total of 40,000 €. Flare cost is 100,000 €. Therefore, total additional cost is 140,000 €.

Compression energy is calculated using the isentropic compression equation:

$$W = \frac{ZRT}{M} \cdot \frac{k}{k-1} \left[\left(\frac{P_2}{P_1} \right)^{\frac{k-1}{k}} - 1 \right] \quad (4.25)$$

Energy use per m³ is 0.2728 MJ/m³, and energy cost per m³ is 0.01061 €/m³.

$$\text{Total Biogas Grid Cost} = 0.064886 + 0.010608 = 0.075494 \text{ €/m}^3 = 7.549 \text{ €ct/m}^3 \quad (4.26)$$

4.9 Upgrading

Table 4.12: Upgrading unit parameters.

Parameter	Value
Depreciation period	12 years
Interest rate	7%
O&M fraction	5%
Green gas capacity	1200 m ³ /h
Operating hours	8000 h/year
Annual production	9,600,000 m ³ /year
Investment	2,764,000 €
Capital Recovery Factor	0.1128
Annualised CAPEX	311,779 €/year
O&M cost	138,200 €/year
Specific energy consumption	0.408 kWh/m ³
Electricity price	0.14 €/kWh
Annual energy cost	548,352 €/year
Total annual upgrading cost	998,331 €/year
Upgrading cost	10.399 €ct/m³

$$\text{CRF} = \frac{r(1+r)^n}{(1+r)^n - 1} = \frac{0.07 \times 1.07^{12}}{1.07^{12} - 1} = 0.1128 \quad (4.27)$$

4.10 Injection

$$\text{Total investment} = 188,500 \text{ €} \quad (4.28)$$

$$\text{Annual depreciation} = \frac{188,500}{12} = 15,708 \text{ €/year} \quad (4.29)$$

$$\text{Annual interest} = 188,500 \times 0.07 = 13,195 \text{ €/year} \quad (4.30)$$

$$\text{Annual OPEX} = 188,500 \times 0.05 = 9,425 \text{ €/year} \quad (4.31)$$

$$\text{Total annual cost} = 15,708 + 13,195 + 9,425 = 38,328 \text{ €/year} \quad (4.32)$$

$$\text{Injection cost} = \frac{38,328}{9,600,000} \times 100 = 0.399 \text{ €ct/m}^3 \quad (4.33)$$

4.11 Final Aggregation

Table 4.13: Components included in total cost and energy aggregation.

Total Cost Components	Total Energy Components
Co-substrate	Digester
Digester	Transport
Transport	Digestate
Grid	Compression
Upgrading	Upgrading
Injection	—

Chapter 5

Comparison Between Kota and Bikaner

5.1 Introduction

The availability of biomass, agricultural output, and transport infrastructure are factors that strongly influence biogas production system performance. This chapter compares two locations in Rajasthan—Kota and Bikaner—to identify critical parameters driving system effectiveness and cost, and to express how geographic diversity affects overall viability.

5.2 Selection of Study Locations

5.2.1 Livestock Distribution

Bikaner has a significantly larger cattle population than Kota, providing more manure as feedstock.

5.2.2 Agricultural Characteristics

Kota benefits from better irrigation and agricultural productivity, particularly for maize. Bikaner's dry climate limits certain crop cultivation; it is more dependent on livestock than crops.

5.2.3 Transport Conditions

Transport distances in Bikaner are longer because biomass sources are dispersed across a much larger district area. These differences directly affect transport energy and cost.

5.3 Key Parameters for Model Updating

5.3.1 Biomass Availability

Table 5.1: Livestock and maize production comparison.

Parameter	Reference	Kota	Bikaner
Number of cattle	4,064	5,149	13,050
Maize production	205,000 ton/year	250,000 ton/year	2 ton/year

5.3.2 Transport Distance

Table 5.2: Transport distance comparison.

Parameter	Reference	Kota	Bikaner
Transport distance	10.64 km	15.06 km	32.59 km

5.3.3 Other Updated Parameters

Table 5.3: Additional model parameters.

Parameter	Reference	Kota	Bikaner
Number of digesters	8	6	6
Annual green gas	9,600,000 m ³ /year	17,021,869 m ³ /year	4,544,901 m ³ /year

Chapter 6

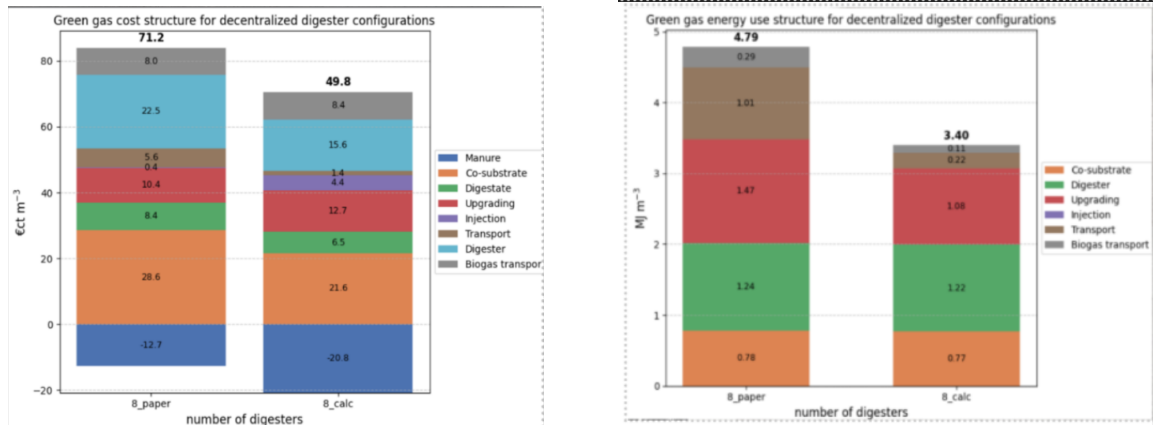
Results and Discussion

6.1 Model Validation

The Excel-based computational model was validated by comparison with the reference study. The comparison showed acceptable accuracy:

- Total production cost from the reference: 71.2 €/t/m³; model result: 49.8 €/t/m³.
- Total energy consumption from the reference: 4.79 MJ/m³; model result: 3.40 MJ/m³.

Despite discrepancies, trends in total cost and energy use were consistent with the reference study, and key design parameters were successfully computed.



(a) Green gas cost structure for decentralised digester configurations.

(b) Green gas energy use structure for decentralised digester configurations.

Figure 6.1: Model validation results comparing reference paper values with model-calculated values.

6.2 Model Improvement and Calibration

The identified discrepancies were addressed by modifying key parameters such as number of cows, co-substrate availability, and investment costs using additional literature references. Justified assumptions were made where data were unavailable, and the model was iteratively calibrated until outputs closely matched reference values.

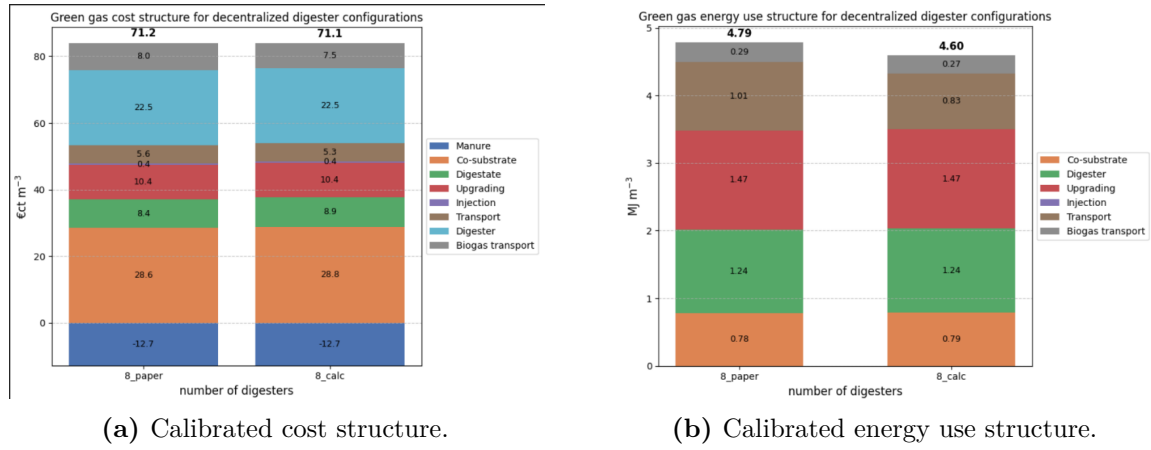


Figure 6.2: Post-calibration results showing improved agreement between reference and model values.

6.3 Cost Analysis: Kota vs. Bikaner

Table 6.1: Production cost comparison between Kota and Bikaner.

Location	Production Cost	Relative Difference
Kota	62.5 €ct/m ³	—
Bikaner	67.9 €ct/m ³	+8.6%

Bikaner is approximately 8.6% more expensive to produce than Kota, primarily due to higher transportation and infrastructural costs.

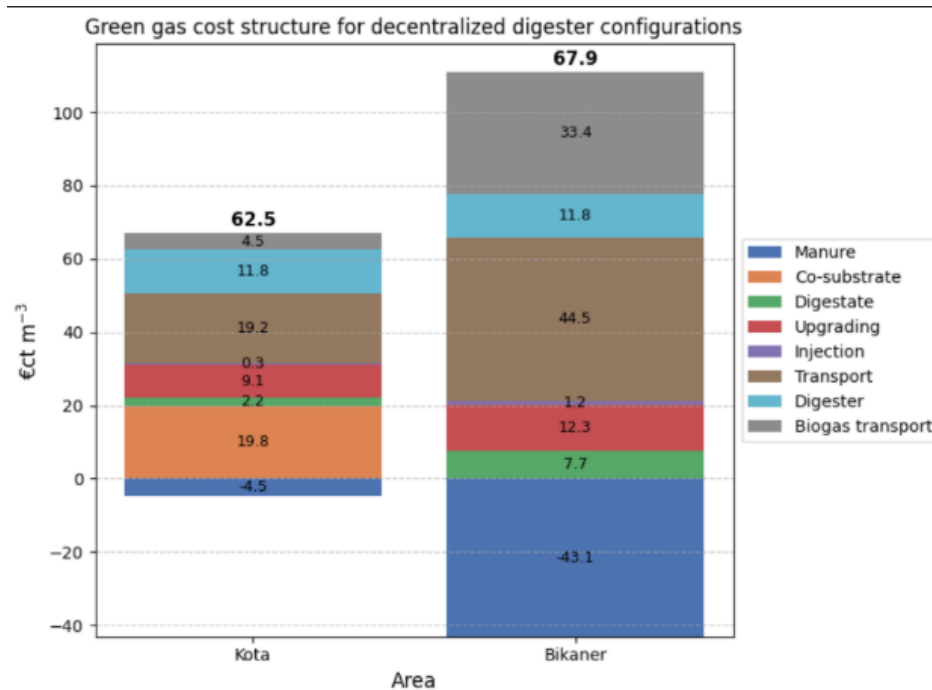


Figure 6.3: Green gas cost structure for Kota and Bikaner.

6.4 Energy Consumption Analysis

Table 6.2: Total energy consumption comparison between Kota and Bikaner.

Location	Energy Consumption	Relative Difference
Kota	6.82 MJ/m ³	–
Bikaner	10.33 MJ/m ³	+51.5%

Bikaner requires 51.5% more energy than Kota. The main contributing factors are transport processes, gas upgrading processes, and digestion. Longer transport distances and higher biomass handling requirements in Bikaner account for the greater energy demand.

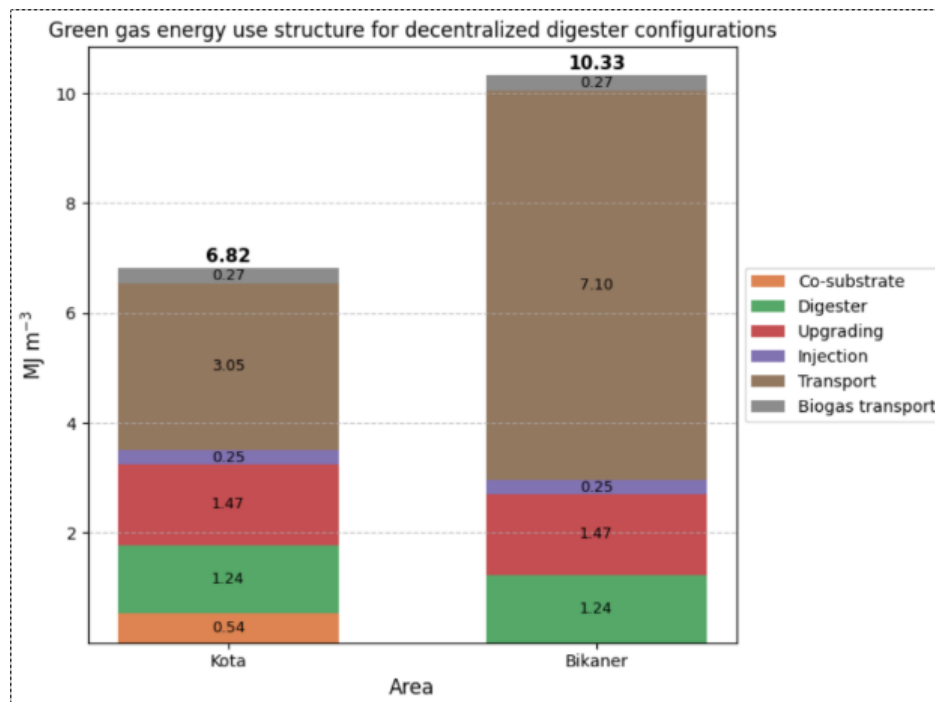


Figure 6.4: Green gas energy use structure for Kota and Bikaner.

6.5 Effect of Geographic and Resource Parameters

Regional characteristics that directly affect system performance include:

- Availability of biomass, including manure and crop residues.
- Transport distance.
- Number of cattle.
- Number of biogas plants and infrastructure requirements.

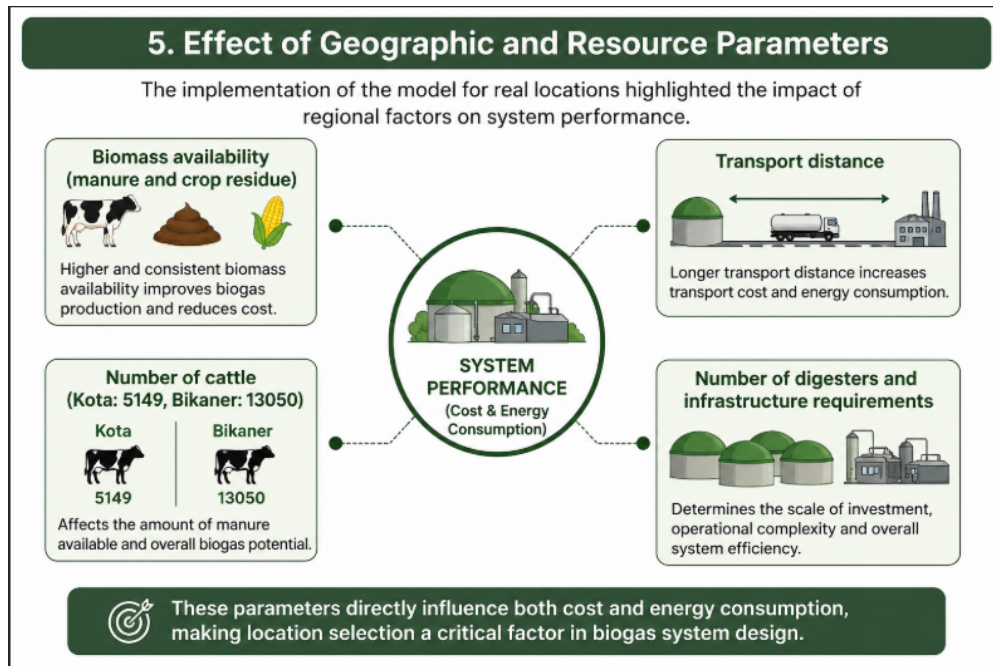


Figure 6.5: Effect of geographic and resource parameters on cost and energy performance.

6.6 Overall Interpretation of Results

Based on the economic and energetic analysis:

- Kota outperforms Bikaner on both economic and energetic metrics.
- Shorter transportation distances in Kota result in lower economic and energetic impacts.
- Superior system integration in Kota improves overall process feasibility.

Therefore, decentralised biogas production with centralised upgrading and grid injection is more feasible in Kota than in Bikaner.

Chapter 7

Conclusion

This study investigated the viability of decentralised biogas production followed by centralised upgrading and grid injection using a computational Excel model. The model was constructed and validated to estimate costs and energy performance.

The comparison between Kota and Bikaner reveals that geographic and resource distribution parameters significantly influence system performance. Kota achieves a total cost of 62.5 €/t/m³ and energy consumption of 6.82 MJ/m³, outperforming Bikaner in both metrics.

The higher cost of 8.6% and higher energy consumption of 51.5% in Bikaner are attributed to greater transportation distances, higher infrastructure demands, and less favourable resource distribution. Transport, upgrading, and digestion operations were identified as the primary energy-consuming components.

On balance, Kota is the preferred location in terms of total cost and energy savings, owing to shorter transportation requirements and a more favourable feedstock distribution.

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